- Navigation ~~&~~ GPS
  Providing precise location data for various systems

- Earth observation & weather
  monitoring weather patterns, disaster management and climate change

- Telemedicine & Education
  Connecting remote clinics to hospitals and delivering e-learning.

Key industry satellite service products

- Satellite Communication services (SATCOM):
  High through put broadband, VSAT ~~for~~ (very small aperture Terminal) for enterprise connectivity and satellite IoT for asset tracking.

- mobile satellite services (MSS):
  Voice, data, and tracking services for maritime, aero-mobile and land-mobile users.

- Broad casting services:
  Direct-to-Home (DTH) satellite TV, radio and content distribution for media

- Earth Observation & Imaging:
  High resolution imagery for monitoring climate agricultural health, disaster management, and urban planning.

- Navigation & positioning:
GNSS (global navigation satellite system) services for precise timing and location data

Advantages

Reliable:
less susceptible to local interference or ground-based issues, offering stable links for critical services

Scalability:
networks can easily expand capacity and coverage by adding more satellites

versatility:
Supports diverse applications, including internet navigation, weather monitoring, military comms and mobile banking

broadcasting:
Excellent for one to many services like TV & radio, efficiently reaching millions simultaneously

mobility:
Supports communication for moving platforms like ships, aircraft, and vehicles, ensuring connectivity on the go

Cost - Effective for distance:
Can be cheaper than terrestrial networks for connecting extremely long distances or vast sparsely populated areas.

Quick deployment
system can be set up much faster than laying fiber optic cables, ideal for temporary needs or new areas.

Disadvantages

- High Initial Investment:
Building, launching and insuring satellites is extremely expensive
-
- Difficult Repairs & limited lifespan:
once in orbit, repairs are complex and costly and satellite eventually need replacement.

- Latency / Propagation Delay:
Signals travel long distances causing noticeable delays which can affect realtime communication like phone calls.

- Weather Dependency:
Heavy rain, storms and atmospheric conditions can weaken or block signals

- Interference:
Susceptible to interference from other satellites, solar flares, and other electronic signals

- limited Bandwidth:
Available bandwidth can be restricted, affecting data rates

- High Power / Large Antennas:
Requires more powerful ground equipment and large antennas.

Operational challenges:

- orbit Congestion
Increasing number of satellites can lead to frequency congestion and collisions

- Environmental Impact
Launching satellites consumes non-renewable resources.

- Positioning & stability
Requires precise control to maintain orbit and antenna alignment

-

Comparison to Fiber:
Fiber optics offers lower latency, higher speeds, and better reliability for dense areas, making satellite communication less competitive for terrestrial broadband but vital for remote coverage.

What is the Internet

The internet is a global network of interconnected computer networks that allows devices all over the world to communicate with each other. it connects millions of computers, servers, mobile phones and smart devices into one massive communication system

- The Internet uses a set of standard communication protocols called TCP/IP (Transmission Control protocol / Internet Protocol). These protocols define how data is broken into packets, transmitted accross networks and reassembled at destination
- one of the main features of the internet is end to end connectivity across heterogeneous networks. This means that different types of networks can all communicate seamlessly using the same protocols.

Basic Network Components.

1 End devices
End devices are the devices used by users to send and receive data. These include
- computers and laptops
- smartphones and tablets
- servers and smart devices
They are called end devices because they are the starting and ending points of communicatio

2 Networking devices
Networking devices connect end devices and manage data flow within and between networks.

- Routers :
Direct data packets between different networks

- Switches:
Connects multiple devices within the same local network and forward data to the correct device

- Gateways:
Act as translators between different types of networks or ~~people~~ protocols.

These devices ensure that data reaches the correct destination efficiently

3 Transmission media

Transmission media refers to the physical or wireless paths through which data travels.

- optical fiber:
uses light signals, very high speed, long distance.
- Copper cables:
such as twisted pair or coaxial, used in LANs
- wireless:
uses radio waves, used in WiFi and mobile networks

4 Address system
Every device on the Internet is identified by an IP address (Internet protocol address).
This address works like a postal address, helping the network know:
- where the data is coming from
- where the data should be delivered

without IP addresses, data routing on the Internet would not be possible

Internet Architecture
Internet architecture describes how computers and networks are organized and interact to provide service over the internet. the main architectural models are

1 <u>client - server Architecture</u>
In client - server architecture, one computer acts as a server and provides services while other computer acts as clients and request those services.

Features:
- The client sends a request (eg open a webpage)
- The server processes the request and sends a response
- The server controls data and resource.

example: web browsing, Email services etc

2 <u>Peer - to - Peer (P2P) Architecture</u>
In peer to peer architecture all computers are equal and acts as both client and server.

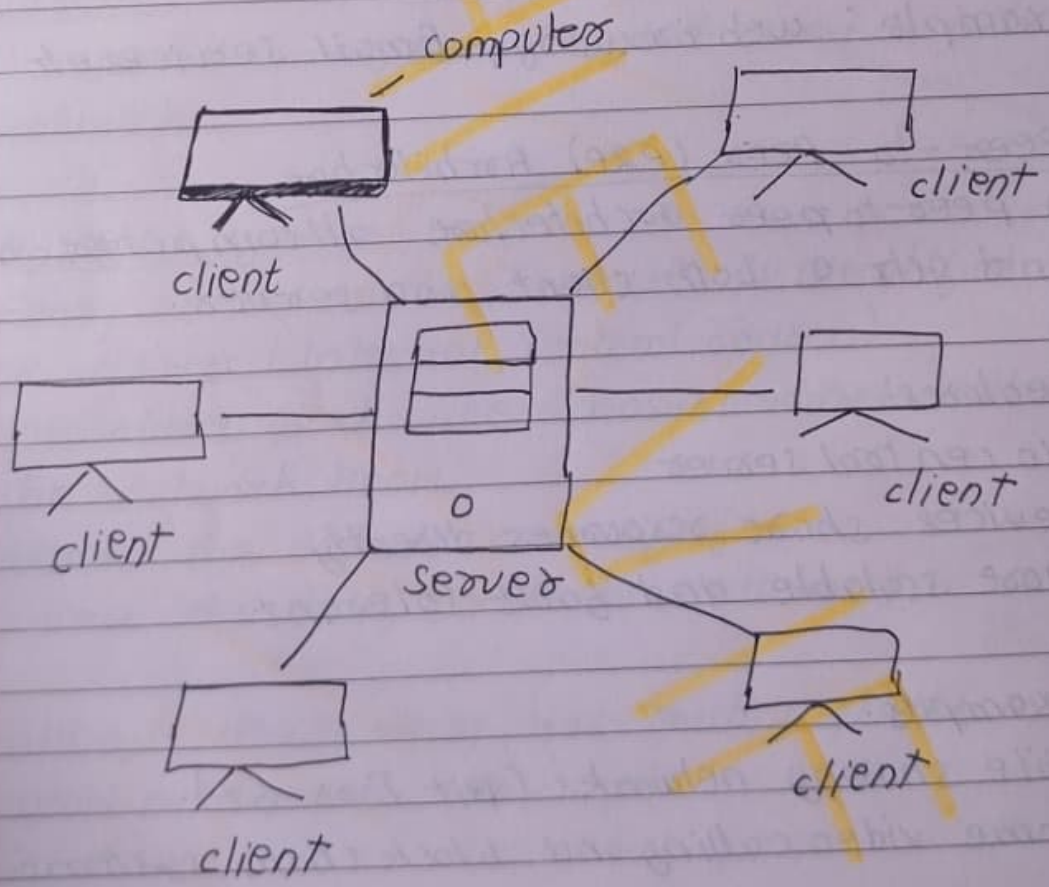
Features:
- No central server
- Devices share resources directly
- More scalable and fault-tolerant

Examples:
- File sharing networks (Bit Torrent).
- Some video calling and block chain systems.

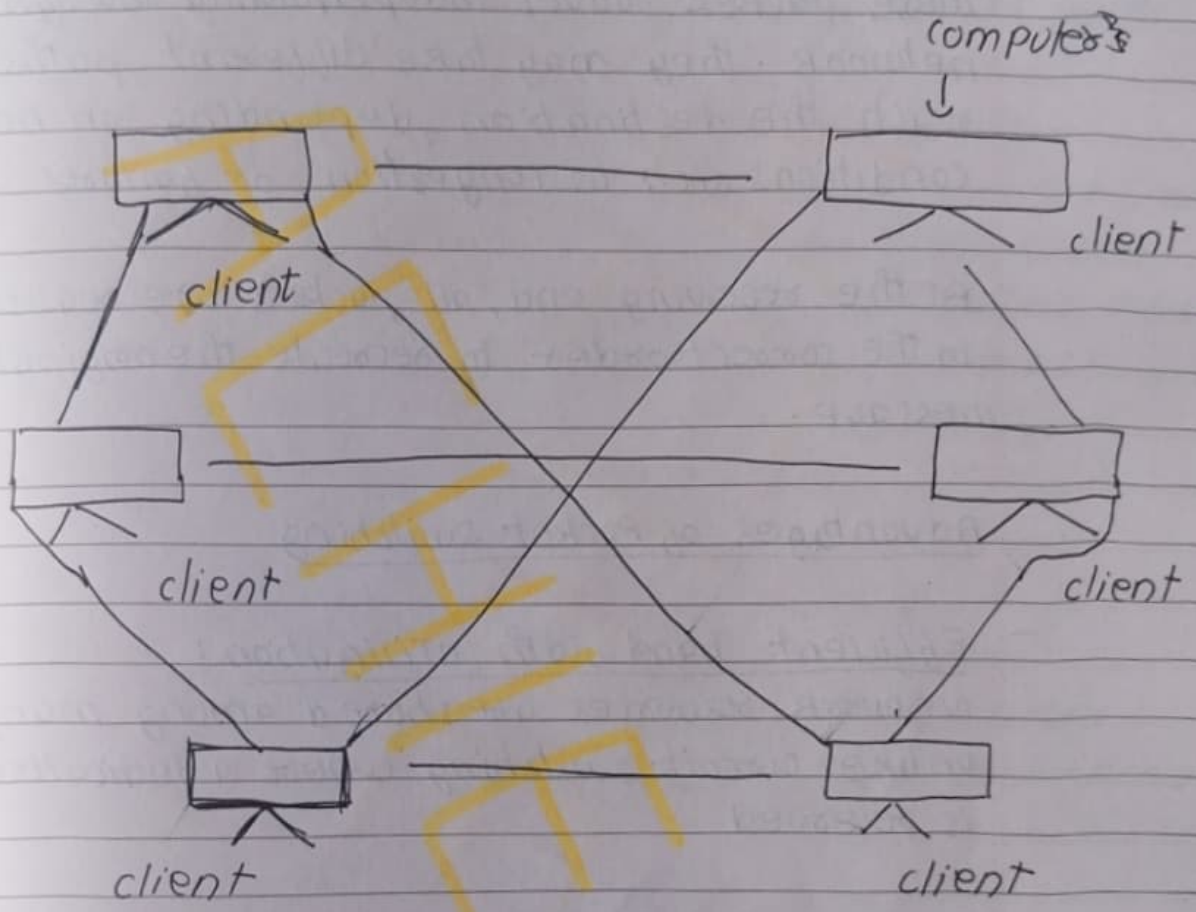
3 <u>Distributed network model</u>
In the distributed network model, data and services are spread across multiple servers and locations instead of one central point.

Features:
- High reliability and availability
- load is shared among many nodes
- Failure of one node does not stop the systems.

Examples:
- cloud computing platforms.
- Content delivery networks like Netflix or Youtube.



Client server Architecture



P2P Architecture

Packet switching Concept

Packet switching is a method of data transmission in which information is divided into small units called packets before being sent over a network

Each packet contains:
- A part of the actual data
- Source and destination addresses
- sequence information

These packets travel independently through the network. They may take different paths to reach the destination, depending on network conditions such as congestion or failures.

At the receiving end, all packets are reassembled in the correct order to recreate the original message.

Advantages of Packet switching

- Efficient bandwidth utilization:
Network resources are shared among many users unlike circuit switching where a dedicated path is reserved

- Better reliability
if one path fails, packets can be ~~re~~ rerouted through another path

- Supports multiple services
suitable for data, voice, video and multimedia communication.

OSI model
OSI stands for open systems ~~intercommuni~~ interconnection. It is a conceptual reference model developed by the International organization for standardization (ISO) in 1984. The OSI model standardizes network communication by dividing it into seven logical layers, making it easier for

different systems and vendors to communicate with each other

- The OSI model acts as a universal language for computer networking. It explains how data moves from an application on one device, travels through the network and reaches an application on another device
- By dividing communication into layers, the OSI model helps in:
- understanding how networks work
- designing network protocols
- Troubleshooting network problems easily

Each layer has a specific function, starting from physical transmission and ending at user applications.

Seven layers of the OSI model

1- Physical layer:
Deals with cables, connectors, and electrical signals.

2- Data link layer:
Handles error detection and frames

3- Network layers:
manages routing and IP addressing

4- Transport layer
ensures reliable data delivery

5- Session layer
manages sessions between devices.

Teacher's Signature:

6 presentation layer
data formatting and encryption.

7 Application layer
user-facing services like email and web

OSI model

	Data	layer
Host layers	Data	Application, network process to Application
	Data	presentation, data representation & Encryption
	Data	session, interhost communication
	Segments	Transport, End to End connections
media layer	Packets	Network, path Determination and IP
	Frames	Data link, MAC and LIC
	Bits	Physical, media signals and binary transmission

TCP/IP model

- The TCP/IP model is the practical and widely used network model that defines how data is broken into packets, addressed, transmitted, routed and reassembled at the destination over the Internet.
- The TCP/IP stands for Transmission control protocol / Internet protocol, which are the core protocols of the Internet
- Unlike the OSI model, the TCP/IP model has 4 layers

layers of the TCP/IP model

1 Application layer

This layer provides services directly to the user and applications

example: HTTP, FTP, SMTP

Functions

- Data formatting and communication with user software
- supports network applications.

2 Transport layer

This layer ensures end-to-end communication between devices

Functions

- Breaks data into segments
- Ensures reliability and error control

Protocols

TCP

UDP

3 Internet layer
This layer handles logical addressing and routing of data packets

Functions:
- Assigns IP addresses
- Finds the best path for packets

Protocols:
- IP
- ICMP

4 Network Access layer
This layer deals with physical transmission of data over the network

Functions:
- Convert digital data into signals
- Handles hardware addressing

Examples
Ethernet, Wi-Fi

Difference between OSI model and TCP/IP model